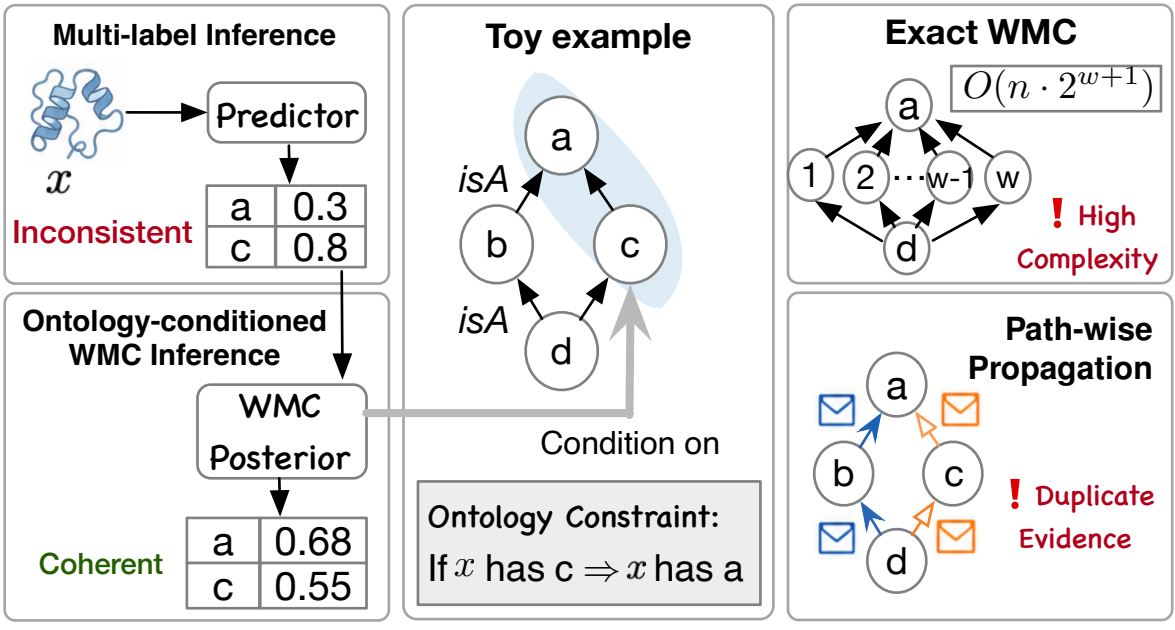

Multi-label Inference
Predictor
x
Inconsistent
a 0.3
c 0.8
Ontology-conditioned WMC Inference
WMC Posterior
Coherent
a 0.68
c 0.55
Toy example
isA
isA
a
b
c
d
Condition on
Ontology Constraint:
If x has c $\Rightarrow$ x has a
Exact WMC
$O(n \cdot 2^{w+1})$
1
2
w-1
w
! High Complexity
Path-wise Propagation
! Duplicate Evidence

(a) (b) (c)